Question No: 24 (Marks: 10)

Evaluate the following limit Q # 1 Whether the given lines are parallel, perpendicular or none of these?

$\frac{1}{2}(y-1)=x-3$ *and* $8-2y=x+7$

Solution:

$L_1: \frac{1}{2}(y-1)=x-3$

$L_2: 8-2y=x+7$

First we have to calculate the slope.

$$L_1 : \frac{1}{2}(y-1) = x-3$$

$$y-1 = 2(x-3)$$

$$y = 2x-6+1$$

$$y = 2x-5$$

Comparing it with equation of line y = m x + c

Slope of L_1 = m_1 = 2

$$L_2 : 8-2y = x+7$$

$$-2y = x+7-8$$

$$-2y = x-1$$

$$y = -\frac{1}{2}x+\frac{1}{2}$$

Comparing it with equation of line y = m x + c

Slope of L_2 = m_2 = - 1 / 2

Hence the given two lines are perpendicular because $m_1 m_2$ = (2) $(-\frac{1}{2})$ = -1

Q # 2

Let

$$f(x) = \frac{x}{x+3} \quad \text{and} \quad g(x) = x^2$$

Find whether $(f \circ g)(x)$ *and* $(g \circ f)(x)$ **Solution:**

are equal or not?

Here

$f(x) = \frac{x}{x+3}$ and $g(x) = x^2$

$(f \circ g)(x) = f\ (\ g\ (x)\ )$

$(f \circ g)(x) = f\ (\ x^2\ )$

$(f \circ g)(x) = \frac{x^2}{x^2+3}$

Also

$(g \circ f\ )(x) = g\ (\ f\ (\ x\ )\ )$

$(g \circ f\ )(x) = g\ (\ \frac{x}{x+3}\ )$

$(g \circ f\ )(x) = \frac{x^2}{(\ x+3\ )^2}$

$(g \circ f\ )(x) = \frac{x^2}{x^{\ 2}+6x+9}$

Hence $(f \circ g)(x) \neq (g \circ f\ )(x)$

Q # 3 Determine whether the equation represents a circle, if the equation represents a circle, find the center and radius?

$x^2 + 2x + y^2 - 4y = 5$

Solution:

First, group the x-terms, group the y-terms, and take the

constant to the right side:

$$(x^2 + 2x) + (y^2 - 4y) = 5$$

$$x^2+2x +1+y^2-4y +4=5+1+4$$

Then write the left side as squares and add up the right side and you get

$$(x+1)^2+(y-2)^2=10,$$

and now you can find the center and radius and graph it. Here the center would be (-1,2) and the radius would be $\sqrt{10}$. So to graph it, all you need to do is find the point (-1,2) and then plot the points $\sqrt{10}$ up, $\sqrt{10}$ down, $\sqrt{10}$ to the right, and $\sqrt{10}$ to the left of it, and draw the circle through them.

Q # 4 Solve the inequality

$$-6 \le \frac{4-2x}{3} < 1$$

Solution:

$$-6 \le \frac{4-2x}{3} < 1$$

multiply by 3, *we get*

$$-18 \le 4 - 2x < 3$$

subtracting by 4

$$-22 \le -2x < -1$$

divide by -2

$$11 \ge x > \frac{1}{2}$$

or $\frac{1}{2} < x \le 11$

Hence the required solution is

$$\left(\frac{1}{2}, 11\right]$$

Q # 1 Whether the given lines are parallel, perpendicular or none of these?

$\frac{1}{2}(y-1)=x-3$ *and* $8-2y=x+7$

Solution:

$L_1:\frac{1}{2}(y-1)=x-3$

$L_2: 8-2y=x+7$

First we have to calculate the slope.

$L_1:\frac{1}{2}(y-1)=x-3$

$y-1=2(x-3)$

$y=2x-6+1$

$y=2x-5$

Comparing it with equation of line y = m x + c

Slope of $L_1 = m_1 = 2$

$L_2: 8-2y=x+7$

$-2y=x+7-8$

$-2y=x-1$

$y=-\frac{1}{2}x+\frac{1}{2}$

Comparing it with equation of line $y = m x + c$

Slope of $L_2 = m_2 = -1/2$

Hence the given two lines are perpendicular because $m_1 m_2 = (2)(-\frac{1}{2}) = -1$

Q # 2

Let

$$f(x) = \frac{x}{x+3} \quad \text{and} \quad g(x) = x^2$$

Find whether $(f \circ g)(x)$ *and* $(g \circ f)(x)$ are equal or not?

Solution:

Here

$$f(x) = \frac{x}{x+3} \quad \text{and} \quad g(x) = x^2$$

$$(f \circ g)(x) = f\,(g\,(x))$$
$$(f \circ g)(x) = f\,(x^2)$$
$$(f \circ g)(x) = \frac{x^2}{x^2+3}$$

Also

$$(g \circ f)(x) = g(f(x))$$

$$(g \circ f)(x) = g\left(\frac{x}{x+3}\right)$$

$$(g \circ f)(x) = \frac{x^2}{(x+3)^2}$$

$$(g \circ f)(x) = \frac{x^2}{x^2 + 6x + 9}$$

Hence $(f \circ g)(x) \neq (g \circ f)(x)$

Q # 3 Determine whether the equation represents a circle, if the equation represents a circle, find the center and radius?

$x^2 + 2x + y^2 - 4y = 5$

Solution:

First, group the x-terms, group the y-terms, and take the constant to the right side:

$$(x^2 + 2x) + (y^2 - 4y) = 5$$

$$x^2+2x+1+y^2-4y+4=5+1+4$$

Then write the left side as squares and add up the right side and you get

$$(x+1)^2+(y-2)^2=10,$$

and now you can find the center and radius and graph it. Here the center would be (-1,2) and the radius would be $\sqrt{10}$. So to graph it, all you need to do is find the point (-1,2) and then plot the points $\sqrt{10}$ up,

$\sqrt{10}$ down, $\sqrt{10}$ to the right, and $\sqrt{10}$ to the left of it, and draw the circle through them.

Q # 4 Solve the inequality

$$-6 \le \frac{4-2x}{3} < 1$$

Solution:

$$-6 \le \frac{4-2x}{3} < 1$$

multiply by 3, *we get*

$$-18 \le 4 - 2x < 3$$

subtracting by 4

$$-22 \le -2x < -1$$

divide by -2

$$11 \ge x > \frac{1}{2}$$

or $\frac{1}{2} < x \le 11$

Hence the required solution is

$$\left(\frac{1}{2}, 11\right]$$

Q # 1 Whether the given lines are parallel, perpendicular or none of these?

$\frac{1}{2}(y-1) = x-3$ *and* $8-2y = x+7$

Solution:

$$L_1 : \frac{1}{2}(y-1) = x-3$$
$$L_2 : \quad 8-2y = x+7$$

First we have to calculate the slope.

$$L_1 : \frac{1}{2}(y-1) = x-3$$
$$y-1 = 2(x-3)$$
$$y = 2x-6+1$$
$$y = 2x-5$$

Comparing it with equation of line y = m x + c

Slope of L_1 = m_1 = 2

$$L_2 : 8-2y = x+7$$
$$-2y = x+7-8$$
$$-2y = x-1$$
$$y = -\frac{1}{2}x + \frac{1}{2}$$

Comparing it with equation of line y = m x + c

Slope of L_2 = m_2 = - 1 / 2

Hence the given two lines are perpendicular because $m_1 m_2$ = (2) $(-\frac{1}{2})$ = -1

Q # 2

Let

$f(x) = \frac{x}{x+3}$ and $g(x) = x^2$

Find whether $(f \circ g)(x)$ *and* $(g \circ f)(x)$ are equal or not?

Solution:

Here

$f(x) = \frac{x}{x+3}$ and $g(x) = x^2$

$(f \circ g)(x) = f\ (g\ (x))$

$(f \circ g)(x) = f\ (x^2)$

$(f \circ g)(x) = \frac{x^2}{x^2+3}$

Also

$(g \circ f)(x) = g\ (f\ (x))$

$(g \circ f)(x) = g\ (\frac{x}{x+3})$

$(g \circ f)(x) = \frac{x^2}{(x+3)^2}$

$(g \circ f)(x) = \frac{x^2}{x^2+6x+9}$

Hence $(f \circ g)(x) \neq (g \circ f)(x)$

Q # 3 Determine whether the equation represents a circle, if

the equation represents a circle, find the center and radius?

$x^2 + 2x + y^2 - 4y = 5$

Solution:

First, group the x-terms, group the y-terms, and take the constant to the right side:

$$(x^2 + 2x) + (y^2 - 4y) = 5$$

$$x^2+2x +1+y^2-4y +4=5+1+4$$

Then write the left side as squares and add up the right side and you get

$$(x+1)^2+(y-2)^2=10,$$

and now you can find the center and radius and graph it. Here the center would be (-1,2) and the radius would be $\sqrt{10}$. So to graph it, all you need to do is find the point (-1,2) and then plot the points $\sqrt{10}$ up, $\sqrt{10}$ down, $\sqrt{10}$ to the right, and $\sqrt{10}$ to the left of it, and draw the circle through them.

Q # 4 Solve the inequality

$$-6 \le \frac{4-2x}{3} < 1$$

Solution:

$$-6 \le \frac{4-2x}{3} < 1$$

multiply by 3, *we get*

$-18 \le 4 - 2x < 3$

subtracting by 4

$-22 \le -2x < -1$

divide by -2

$$11 \ge x > \frac{1}{2}$$

or $\frac{1}{2} < x \le 11$

Hence the required solution is

$\left(\frac{1}{2}, 11\right]$

$$\lim_{y \to -2} g(y) \quad \textit{where,} \quad g(y) = \begin{cases} y^2 + 5 & \textit{if } y < -2 \\ 3 - 3y & \textit{if } y \ge -2 \end{cases}$$

ALL IN ONE Mega File
MTH101 Midterm PAPERS,
MCQz & subjective
Created BY Farhan & Ali
BS (cs) 3rd sem
Hackers Group
Mandi Bahauddin
Remember us in your prayers

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

Mindhacker124@gmail.com

Hearthacker124@gmail.com

Question No.1

Use implicit differentiation to find dy/dx if

$$x^2 = \frac{x+y}{x-y}$$

Solution:

$$x^2 = \frac{x+y}{x-y}$$

$$\frac{d(x^2)}{dx} = \frac{(x-y)\left(\frac{d(x+y)}{dx}\right) - (x+y)\left(\frac{d(x-y)}{dx}\right)}{(x-y)^2}$$

$$2x = \frac{(x-y)\left(1+\frac{dy}{dx}\right) - (x+y)\left(1-\frac{dy}{dx}\right)}{(x-y)^2}$$

$$2x = \frac{x-y+x\frac{dy}{dx} - y\frac{dy}{dx} - x - y + x\frac{dy}{dx} + y\frac{dy}{dx}}{(x-y)^2}$$

$$2x = \frac{-2y + 2x\frac{dy}{dx}}{(x-y)^2}$$

$$2x(x-y)^2 = -2y + 2x\frac{dy}{dx}$$

$$x(x-y)^2 = -y + x\frac{dy}{dx}$$

$$x(x-y)^2 + y = x\frac{dy}{dx}$$

$$\frac{x(x-y)^2 + y}{x} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x(x-y)^2 + y}{x}$$

Some other method can be used to solve this.

Question No. 2.

Find the slope of the tangent line to the given curve at the specified point

$$2(x^2 + y^2)^2 = 25(x^2 - y^2); \qquad (3,1)$$

Solution:

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$

differentiating with respect to 'x'

$$2 \times 2(x^2 + y^2)\frac{d(x^2 + y^2)}{} = 25\frac{d(x^2 - y^2)}{}$$

$$4(x^2 + y^2)(2x + 2y\frac{dy}{dx}) = 25(2x - 2y\frac{dy}{dx})$$

$$4(x^2 + y^2).2(x + y\frac{dy}{dx}) = 2(25x - 25y\frac{dy}{dx})$$

$$8(x^2 + y^2)(x + y\frac{dy}{dx}) = 2(25x - 25y\frac{dy}{dx})$$

$$4(x^2 + y^2)(x + y\frac{dy}{dx}) = (25x - 25y\frac{dy}{dx})$$

$$4x(x^2 + y^2) + 4y(x^2 + y^2)\frac{dy}{dx} = 25x - 25y\frac{dy}{dx}$$

$$4y(x^2 + y^2)\frac{dy}{dx} + 25y\frac{dy}{dx} = -4x(x^2 + y^2) + 25x$$

$$4y(x^2 + y^2)\frac{dy}{dx} + 25y\frac{dy}{dx} = -x(4x^2 + 4y^2) + 25x$$

$$y(4x^2 + 4y^2)\frac{dy}{dx} + 25y\frac{dy}{dx} = x(-4x^2 - 4y^2 + 25)$$

$$y(4x^2 + 4y^2 + 25)\frac{dy}{dx} = x(-4x^2 - 4y^2 + 25)$$

$$\frac{dy}{dx} = \frac{x(-4x^2 - 4y^2 + 25)}{y(4x^2 + 4y^2 + 25)}$$

At point (3,1)

$$\left.\frac{dy}{dx}\right|_{(3,1)} = \frac{x(-4x^2 - 4y^2 + 25)}{y(4x^2 + 4y^2 + 25)}$$

$$\left.\frac{dy}{dx}\right|_{(3,1)} = \frac{3(-4\times 3^2 - 4\times 1^2 + 25)}{1(4\times 3^2 + 4\times 1^2 + 25)}$$

$$= \frac{3(-36 - 4 + 25)}{36 + 4 + 25}$$

$$\left.\frac{dy}{dx}\right|_{(3,1)} = \frac{-3\times 15}{40 + 25}$$

$$= \frac{-45}{65}$$

$$= \frac{-9}{13}$$

Question No. 3

A 10 –f t ladder is leaning against a wall. If the top of the ladder slips down the wall at the rate of 2- ft/sec, how fast will the foot be moving away from the wall when the top is 6 ft above the ground?

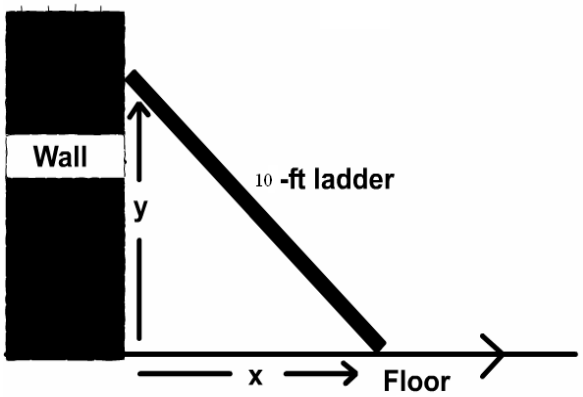
Solution:

Let

x = Distance in feet between wall and foot of the ladder

y = Distance in feet between floor and top of the ladder

t = number of seconds after the ladder starts to slip.



$$\frac{dy}{dt} = -2\ ft/\sec$$

It is negaive because y is decreasing as time increases, since the ladder is slipping down the wall.

$\frac{dx}{dt} = ?$

Using Pythagoras Theorem,

$x^2 + y^2 = 10^2$

differentiating w.r.t 't'

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

$\frac{dx}{dt} = -\frac{y}{x}\frac{dy}{dt}$

when $y = 6\,ft$

$\frac{dx}{dt} = -\frac{6}{x}\frac{dy}{dt}$ .....(i)

Whenthetopis 6 *ft abovethe ground*

$x^2 + 6^2 = 10^2$

$x^2 = 100 - 36$

$x^2 = 64$

$x = 8$

put in (i) *we get*

$\frac{dx}{dt} = -\frac{6}{8}\frac{dy}{dt}$

$\frac{dx}{dt} = -\frac{6}{8}(-2)$

$= \frac{12}{8}$

$= \frac{3}{2} ft / \sec$

ALL IN ONE Mega File
MTH101 Midterm PAPERS,
MCQz & subjective
Created BY Farhan & Ali

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

BS(cs)3rdsemHackersGroup

Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

Question No.1 :

Find the relative extreme values of the function

$$f(x) = a \sin x + b \cos x$$

Solution:

Since,

$$f(x) = a \sin x + b \cos x$$

$f'(x) = a\cos x - b\sin x \ldots\ldots\ldots\ldots\ldots (i)$

$put\ f'(x) = 0 \Rightarrow a\cos x - b\sin x = 0$

$\Rightarrow a\cos x = b\sin x$

$\Rightarrow \frac{a}{b} = \tan x$

$\Rightarrow \tan x = \frac{a}{b} \Rightarrow x = \tan^{-1}\left(\frac{a}{b}\right)$

By Pythagorustheorem

$\Rightarrow \sin x = \frac{a}{\pm\sqrt{a^2+b^2}}, \quad \cos x = \frac{b}{\pm\sqrt{a^2+b^2}}$

Now, taking sec *ond derivative*

$f''(x) = -a\sin x - b\cos x \ldots\ldots\ldots\ldots\ldots\ldots (ii)$

$put \quad \sin x = \frac{a}{\sqrt{a^2+b^2}}, \quad \cos x = \frac{b}{\sqrt{a^2+b^2}} in\ (ii)$

$f''(x) = -a\frac{a}{\sqrt{a^2+b^2}} - b\frac{b}{\sqrt{a^2+b^2}}$

$f''(x) = -\left(\frac{a^2}{\sqrt{a^2+b^2}} + \frac{b^2}{\sqrt{a^2+b^2}}\right) = -\sqrt{a^2+b^2} < 0$

So $f(x)$ *has relative* max *ima at* $\sin x = \frac{a}{\sqrt{a^2+b^2}}$ *and* $\cos x = \frac{b}{\sqrt{a^2+b^2}}$

and sin *ceboth* sin *and* cos *are positvein first quadrant so we say* f *has relative* max *ima at*

$x = \frac{\pi}{2} - \tan^{-1}\left(\frac{a}{b}\right)$

$put \quad \sin x = \frac{-a}{\sqrt{a^2+b^2}}, \quad \cos x = \frac{-b}{\sqrt{a^2+b^2}} in\ (ii)$

$f''(x) = a\frac{a}{\sqrt{a^2+b^2}} + b\frac{b}{\sqrt{a^2+b^2}}$

$f''(x) = \left(\frac{a^2}{\sqrt{a^2+b^2}} + \frac{b^2}{\sqrt{a^2+b^2}}\right) = \sqrt{a^2+b^2} > 0$

So $f(x)$ *has relative* min *ima at* $\sin x = \frac{-a}{\sqrt{a^2+b^2}}$ *and* $\cos x = \frac{-b}{\sqrt{a^2+b^2}}$

and sin *ceboth* sin *and* cos *are positveinthird quadrant so we say* f *has relative* min *ima at* $x = \frac{3\pi}{2} - \tan^{-1}(\frac{a}{b})$

Question No.2:

Let

$$f(x) = \begin{cases} x^2, & \text{if } x \le 1 \\ x, & \text{if } x > 1 \end{cases}$$

Does Mean Value Theorem hold for f on $[\frac{1}{2}, 2]$?

Solution:

To check whether f satisfies M.V.T

I. Clearly f is continuous on $[\frac{1}{2}, 2]$.

II. To check f is differentiable on $]\frac{1}{2}, 2[$

Wecheck whether $f'(1)$ *exists.*

$$L.H.Limt\ f'(1) = \lim_{x \to 1-0} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \to 1-0} \frac{x^2 - 1}{x - 1}$$

$$= \lim_{x \to 1-0} x + 1 = 2$$

$$R.H.Limtf'(1) = \lim_{x \to 1+0} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \to 1-0} \frac{x - 1}{x - 1} = 1$$

Thus $L.H.Limt\ f'(1) \neq R.H.Limtf'(1)$.
Therefore, $f'(1)$ *does not exist and the M.V.T.does not hold on* $[\frac{1}{2}, 2]$.

Question No.3 :

Integrate the following

(i) $\frac{xa^{x^2}}{x^2}$ (ii) $\frac{\ln x}{x}$

Solution:

(i) $\int \frac{xa^{x^2}}{x^2} dx$

$put\ x^2 = t \Rightarrow 2xdx = dt \Rightarrow xdx = \frac{dt}{2}$

$$\int \frac{a^t dt}{2t} = \frac{1}{2}\int \frac{a^t}{t} dt = \frac{1}{2}\int [\frac{1}{t}(1 + t\ln a + \frac{t^2}{2!}(\ln a)^2 + \frac{t^3}{3!}(\ln a)^3 + ...)]$$

$$(\because a^x = 1 + x\ln a + \frac{(\ln a)^2}{2!} + \frac{(\ln a)^3}{3!} + ...)$$

$$= \frac{1}{2}\int [\frac{1}{t} + \ln a + \frac{t}{2!}(\ln a)^2 + \frac{t^2}{3!}(\ln a)^3 + ...)]$$

$$= \frac{1}{2}(\ln t + t\ln a + \frac{t^2}{2}(\ln a)^2 + \frac{t^3}{4}(\ln a)^3 + ...)$$

$$= \frac{1}{2}(\ln x^2 + x^2 \ln a + \frac{(x^2)^2}{4}(\ln a)^2 + \frac{(x^2)^3}{18}(\ln a)^3 + ...)$$

(ii) $\int \frac{\ln x}{x} dx$

$put\ \ln x = t \Rightarrow \frac{1}{x} dx = dt$

t^2

$$\int t dt = \frac{}{2} + c$$

Byback substitution

$$\int t dt = \frac{(\ln x)^2}{2} + c$$

Question 1

Marks 8

Find area of region enclosed by given curves

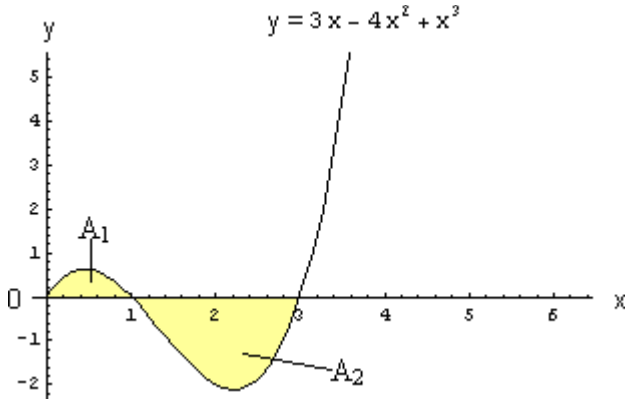
$y = 3x - 4x^2 + x^3$, $y = 0, x = 0, x = 3.$

Solution:

$y = x^3 - 4x^2 + 3x$, $y = 0, x = 0, x = 3.$

y = 0 is the x-axis, x = 0 is y-axis and $x = 3$ is a line parallel to y-axis crossing x-axis at point 3. So we have to find out the area bounded by the curve $y=x^3 -4x^2 +3x$ and the x-axis over the interval [0,3]

Graph of $y = 3x - 4x^2 + x^3$ is below



We have to find the area of shaded region. It is clear from the graph that total area, A, under the curve in interval [0,3] is divided into two regions A_1 and A_2. In [0,1], A_1 is above x-axis and in [1,3], A_2 is below x-axis. So

$$A = A_1 - A_2 = \int_0^3$$

$\int^{1}\left(x^3-4x^2+\right.$ $\int^{3}\left(x^3-4x^2+3x\right)dx$

$\left.3x\right)dx$ $-$

NOTE: Since $A_2 = \int_1^3 \left(x^3 - 4x^2 + 3x\right)dx$ is below x-axis , so we will get negative value of integral. That's why we subtract A_2 from A_1.

Even if we don't have the graph, we can come to this conclusion by first finding the zero of a function that lie between given interval [0,3].

Zero of a function is a value of x which makes a function f(x) equal to zero.

$$x^3 - 4x^2 + 3x = 0$$
$$x(x^2 - 4x + 3) = 0$$
$$x(x^2 - 3x - x + 3) = 0$$

$$x\{x(x-3) - 1(x-3)\} = 0$$
$$x(x-3)(x-1) = 0$$
$$x = 0, 1, 3$$

So zero of a function $\left(x^3 - 4x^2 + 3x\right)$ is 0, 1 and 3

Now 1 lie between our given interval [0, 3], which means that total area is divided into two regions. One in interval [0,1] and other in [1,3]

So

$$A = A_1 - A_2 = \int^{1} \left(x^3 - 4x^2 + 3x\right) dx \; - \; \int^{3} \left(x^3 - 4x^2 + 3x\right) dx$$

$$= \left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right)\Bigg|_0^1 - \left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right)\Bigg|_1^3$$

$$= \frac{1^4}{4} - \frac{4(1)^3}{3} + \frac{3(1)^2}{2} - \left(\frac{0^4}{4} - \frac{4(0)^3}{3} + \frac{3(0)^2}{2} \right) - \left[\frac{3^4}{4} - \frac{4(3)^3}{3} + \frac{3(3)^2}{2} - \left(\frac{1^4}{4} - \frac{4(1)^3}{3} + \frac{3(1)^2}{2} \right) \right]$$

$$= \frac{1}{4} - \frac{4}{3} + \frac{3}{2} - \left[\frac{81}{4} - \frac{108}{3} + \frac{27}{2} - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} \right) \right]$$

$$= \frac{5}{12} + \frac{9}{4} + \frac{5}{12}$$

$$= \frac{37}{12}$$

Question 2

Evaluate

Marks 7

$$\int_0^{\infty} x\, e^{-x^2}\, dx$$

Solution:

$$\int_0^{\infty} x\, e^{-x^2}\, dx$$

$$= \lim_{t \to \infty} \int_0^{t} x\, e^{-x^2}\, dx$$

$$= -\frac{1}{2} \lim_{t \to \infty} \int_0^t (-2x)\, e^{-x^2}\, dx$$

$$= -\frac{1}{2} \lim_{t \to \infty} \left| e^{-x^2} \right|_0^t$$

$$= -\frac{1}{2}[0 - 1] = \frac{1}{2}$$

Question 3 Marks 10

Evaluate

$$\int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)}$$

Solution:

Consider,

$$\frac{1}{(x^2 + a^2)(x^2 + b^2)} = \frac{Ax + B}{x^2 + a^2} + \frac{Cx + D}{x^2 + b^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$\Rightarrow \quad 1 = (Ax + B)(x^2 + b^2) + (Cx + D)(x^2 + a^2)$$

$$\Rightarrow \quad 1 = Ax^3 + Bx^2 + Axb^2 + Bb^2 + Cx^3 + Dx^2 + Cxa^2 + Da^2$$

$$\Rightarrow \quad 1 = (A + C)x^3 + (B + D)x^2 + (Ab^2 + Ca^2)x + (Bb^2 + Da^2)$$

Comparingcoefficients

$$A + C = 0 \quad , \quad B + D = 0$$

$$Ab^2 + Ca^2 = 0 \quad , \quad Bb^2 + Da^2 = 1$$

Here, $C = -A \Rightarrow Ab^2 - Aa^2 = 0$

$\Rightarrow A(b^2 - a^2) = 0 \Rightarrow A = 0$ *if* $b^2 - a^2 \neq 0$

$\Rightarrow C = 0$

Since, $B + D = 0 \quad \Rightarrow \quad D = -B$

thus $\Rightarrow \quad Bb^2 - Ba^2 = 1 \quad \Rightarrow B(b^2 - a^2) = 1 \Rightarrow B = \dfrac{1}{(b^2 - a^2)}$

$$\Rightarrow D = \frac{-1}{(b^2 - a^2)}$$

So (1) *becomes*

$$\frac{1}{(x^2 + a^2)(x^2 + b^2)} = \frac{(0)x + 1/(b^2 - a^2)}{x^2 + a^2} + \frac{(0)x - 1/(b^2 - a^2)}{x^2 + b^2}$$

$$= \frac{1}{(b^2 - a^2)(x^2 + a^2)} - \frac{1}{(b^2 - a^2)(x^2 + b^2)}$$

$$= \frac{1}{b^2 - a^2}\left[\frac{1}{x^2 + a^2} - \frac{1}{x^2 + b^2}\right]$$

int*egrating both sides,*

$$\int_0^\infty \frac{dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{1}{a^2 - b^2}\left[\int_0^\infty \frac{dx}{(x^2 + b^2)} - \int_0^\infty \frac{dx}{(x^2 + a^2)}\right]$$

$$= \frac{1}{a^2 - b^2}\lim_{t\to\infty}\left[\int_0^t \frac{dx}{(x^2 + b^2)} - \int_0^t \frac{dx}{(x^2 + a^2)}\right]$$

$$= \frac{1}{a^2 - b^2}\lim_{t\to\infty}\left[\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right) - \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)\right]_0^t$$

$$= \frac{1}{a^2 - b^2}\lim_{t\to\infty}\left[\frac{1}{b}\tan^{-1}\left(\frac{t}{b}\right) - \frac{1}{a}\tan^{-1}\left(\frac{t}{a}\right)\right]$$

$$= \frac{1}{a^2 - b^2}\left[\frac{1}{b}\cdot\frac{\pi}{2} - \frac{1}{a}\cdot\frac{\pi}{2}\right] = \frac{\pi}{2(a^2 - b^2)}$$

Question No.1 $a^2 - b^2 \; t \to \infty \; \lfloor 2b \quad 2a \rfloor \quad 2ab(a+b)$

Marks 10

Find the volume of the solid that results when the region enclosed by the given curve is revolved about the x-axis

$$y = \sqrt{25 - x^2}, \; y = 3$$

Solution:

We know the volume formula, when we revolve $y = f(x)$ about x-axis, is given by

$$\int_a^b \pi([f(x)]^2 - [g(x)]^2)dx$$

Here given that $y = \sqrt{25 - x^2}$ *and* $y = 3$.

Now to find the initial and final limits of integrations we shall solve both of curves simultaneously to find their points of intersection. On equating both functions we have

$$\sqrt{25 - x^2} = 3$$
$$\Rightarrow \quad 25 - x^2 = 9$$
$$\Rightarrow \quad x^2 = 16$$
$$\Rightarrow \quad x = \pm 4$$
$$\Rightarrow \quad x = -4,\ 4$$

Hence given $a = -4,\ and\ \ b = 4$, so we have

$$\int_a^b \pi([f(x)]^2 - [g(x)]^2)dx = \int_{-4}^4 \pi([\sqrt{25 - x^2}]^2 - [3]^2)dx$$
$$= 2\pi \int_0^4 \left(25 - x^2 - 9\right)dx = 2\pi \int_0^4 \left(16 - x^2\right)dx$$
$$= 2\pi \left[16x - \frac{x^3}{3}\right]_0^4$$
$$= 2\pi \left[64 - 64/3\right]$$
$$= \frac{256\pi}{3}$$

Question No. 2 Marks 8

Use cylindrical shells to find the volume of the solid generated when the region enclosed by the given curves is revolved about the y-axis.

$y=\sqrt{x},\ x=4, x=9, y=0$

Solution:

To find the volume when we revolve around y-axis, we integrate with respect to x i.e.

$$V=\int_a^b 2\pi xf\left(x\right)dx$$

the curve is $y=\sqrt{x}$ *so*

$$V=\int_a^b 2\pi xf\left(x\right)dx=\int_a^b 2\pi x[\sqrt{x}]dx$$

Here $a=4$, *and* $b=9$

Thus

$$V=\int_4^9 2\pi x[\sqrt{x}]dx$$

$$=2\pi[\int_4^9 x^{3/2}dx$$

$$=844\pi\ /5$$

Question No.3

Find the area of the surface generated by revolving the given curve about the y –axis

$x=\sqrt{16-y},\ 0\le y\le 15$

Solution:

Here given that

$$x = g(y) = \sqrt{16 - y}, \qquad 0 \le y \le 15$$

$$\Rightarrow \quad g'(y) = -\frac{1}{2\sqrt{16 - y}}$$

$$\Rightarrow \quad \left(g'(y)\right)^2 = \frac{1}{4(16 - y)}$$

$$\Rightarrow \quad 1 + \left(g'(y)\right)^2 = 1 + \frac{1}{4(16 - y)}$$

$$\Rightarrow \quad = \frac{65 - 4y}{4(16 - y)}$$

$$\Rightarrow \quad \sqrt{1 + \left(g'(y)\right)^2} = \sqrt{\frac{65 - 4y}{4(16 - y)}}$$

Since

$$S = \int_0^{15} 2\pi x \sqrt{1 + \left(g(y)'\right)^2}\, dy$$

So, by above values

$$S = 2\pi \int_0^{15} \left(\sqrt{16 - y}\right)\left(\sqrt{\frac{65 - 4y}{4(16 - y)}}\right) dy$$

$$= (65\sqrt{65} - 5\sqrt{5})\frac{\pi}{6}$$